

PROJECT DOCUMENT – REACH AI

Team Name: Team 12

Team Members:

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1. Title – ReachAI: AI-Based Accessibility Assistant for Short-Height Individuals

ReachAI is an AI-powered web application designed to assist short-height individuals in performing daily activities comfortably. The system helps users determine whether they can physically reach objects such as switches, shelves, ATM panels, and bus handles. The platform provides personalized accessibility suggestions and guidance using intelligent calculations and user input. The goal of the system is to improve independence, safety, and comfort in everyday environments through an easy-to-use digital interface.

2. Problem Statement

To design an AI-based system that identifies challenges faced by short-height individuals in daily activities and provides personalized assistance or solutions to improve their accessibility and comfort.

Short-height individuals frequently experience difficulty interacting with objects placed at higher levels in public and private environments. Many infrastructures are designed for average human height, which creates accessibility barriers. ReachAI aims to solve this problem using AI-based reachability analysis and smart recommendations.

3. Objective

The main objective of ReachAI is to develop a simple and accessible AI assistant that helps users understand whether an object is reachable and provides helpful solutions when it is not.

- Provide a reachability checking system based on user height.
- Assist users in daily activities such as using switches, shelves, ATMs, and public transport handles.
- Offer smart accessibility suggestions.
- Improve independence and confidence of short-height individuals.
- Promote inclusive and accessible technology.

4. Tools Used

- AI Prompt-based website generation
- ChatGPT / OpenAI – Content generation and logic planning
- HTML, CSS, and JavaScript – Web structure and functionality
- Text-to-Speech API – Voice assistant feature
- Responsive UI Design – Mobile and desktop compatibility

Design Features:

- Soft blue gradient background
- Rounded UI cards
- Accessible font sizes
- Mobile-friendly layout
- Simple interface suitable for elderly and differently-abled users

5. Methodology

- Problem Identification – Recognized accessibility challenges faced by short-height individuals in everyday life.
- Planning & Requirements – Decided website features such as homepage, reachability checker, suggestions, and voice assistant.
- Reachability Logic Design – $\text{Reach Height} = 1.25 \times \text{User Height}$.
- Reachability Analysis – Compared reach height with object height entered by the user.
- Smart Suggestion System – Provided appropriate guidance and recommendations.
- Voice Assistant Integration – Implemented text-to-speech to read results aloud.
- Testing & Improvement – Checked responsiveness, usability, and accessibility across devices.

6. System Features (Output)

- Homepage with project introduction
- Reachability Checker with user height and object height input
- Result feedback with colored messages (Green, Yellow, Orange, Red)
- Smart Suggestions section
- Voice assistant button 'Ask ReachAI'
- Mobile-responsive and user-friendly interface
- Accessibility-focused design

7. Result

The ReachAI system successfully demonstrates how AI can be used to improve accessibility in daily life. Users can easily check reachability of objects and receive helpful guidance. The platform increases user independence, reduces difficulty in daily tasks, and spreads awareness about inclusive technology.

8. Conclusion

ReachAI shows that AI-based solutions can significantly improve accessibility for short-height individuals. By combining simple height analysis, smart suggestions, and voice assistance, the system provides a practical and inclusive solution. The project highlights the importance of designing technology that supports people with physical limitations and promotes equal usability for all users.

9. Project URL

<https://reach-high-spaces.lovable.app>

10. GitHub Repository

<https://github.com/RSantosreddy/Reach-AI.git>

https://github.com/anjith6666/Reach-AI_Anjith-rao.git

<https://github.com/NManojkumar23/Reach-AI.git>

https://github.com/reddynitish034-oss/Reach-AI_Nitish-.git

https://github.com/SandeepReddy0129/Reach-Ai_Sandeep.git